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Decentralized Electronic Voting using Blockchain: State-of-the-Art, Limitations, and Research Directions

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Abstract

The emergence of blockchain technology has created transformative opportunities for developing secure, transparent, and trustworthy electronic voting systems. Conventional electronic voting platforms continue to face challenges such as centralization, manipulation of results, lack of transparency, and limited verifiability, which weaken public confidence in electoral integrity. Blockchain introduces a decentralized structure that ensures transparency, traceability, and immutability of voting data while preserving voter anonymity and auditability. This paper presents a detailed review of decentralized electronic voting using blockchain technology and analyzes the current state of research, design principles, and implementation frameworks. It examines the underlying architectural models, cryptographic algorithms, and consensus protocols adopted in contemporary studies and highlights how decentralization strengthens trust among stakeholders. The paper also discusses critical limitations including scalability constraints, latency, energy consumption, coercion resistance, and the complexities of regulatory compliance and user adoption. Through an extensive comparative evaluation of frameworks such as Ethereum, Hyperledger Fabric, Exonum, and Quorum, the study provides a consolidated view of how blockchain can enhance electoral reliability. Finally, it identifies key research directions focusing on lightweight cryptographic schemes, privacy preserving mechanisms, interoperability among distributed ledgers, and integration with artificial intelligence for real time fraud detection. The insights gained from this analysis contribute to advancing secure, transparent, and inclusive democratic processes in the digital era.

Keywords

Blockchain technology, Electronic voting systems, Security and transparency, Decentralization, Cryptographic methods

1. Introduction

The concept of democracy is founded on fair representation, free participation, and transparent electoral processes. Voting systems constitute the foundation of every democratic government, allowing citizens to express their preferences and influence political decisions. Over the past few decades, technological advancement has transformed the traditional methods of voting, replacing paper ballots and manual counting with electronic voting systems designed to increase speed, accuracy, and convenience. Despite these benefits, electronic voting has raised critical concerns regarding data security, transparency, and the potential for manipulation. Centralized architectures remain susceptible to unauthorized access, denial of service attacks, and vote alteration, which can undermine voter confidence and compromise the legitimacy of election outcomes. As Abuidris et al. [1] observed, the absence of transparency and resilience in conventional electronic voting frameworks underscores the need for secure, verifiable, and trustworthy alternatives capable of sustaining democratic integrity in the digital era.

Blockchain technology has emerged as a promising innovation that addresses many of these challenges. By distributing control across a decentralized network, blockchain eliminates dependence on a central authority and provides an immutable public record of all transactions. Each transaction, or in this case each vote, is verified through a consensus mechanism involving multiple network participants. Akritidis et al. [2], [3] highlighted that the The decentralized nature of blockchain, combined with cryptographic verification, ensures that once a vote is recorded and the risks associated with it, it cannot be altered or erased. This property provides unprecedented [36] protection against post-election tampering and strengthens confidence in the reliability of results. Moreover, the transparency of blockchain allows stakeholders to independently verify the voting process while ensuring voter anonymity and confidentiality through encryption and digital signatures.

Unlike traditional systems, where data storage and verification are handled by a central database, blockchain distributes

these responsibilities among nodes in a peer-to-peer network. Every node maintains a synchronized copy of the ledger, and any attempt to alter information requires approval from the majority of nodes. This makes data manipulation practically infeasible and ensures that all voting records are verifiable and auditable. Each entry in the ledger is timestamped and cryptographically linked to previous entries, creating an unbroken and tamper-resistant chain of trust. This immutable record enhances transparency and accountability, allowing public scrutiny without revealing voter identities. In this way, blockchain provides both verifiability and privacy two essential yet often conflicting requirements in electronic voting.

Blockchain also enhances the resilience and accessibility of voting systems. Because data is stored across multiple distributed nodes, the system remains operational even if some nodes fail or are compromised. This redundancy improves fault tolerance and prevents single points of failure that commonly affect centralized systems. Furthermore, blockchain-based e-voting platforms can enable remote participation, expanding access to citizens who are geographically distant, physically challenged, or living abroad. Such inclusivity aligns with the democratic principle of universal suffrage and contributes to higher voter turnout.

In addition to integrity and accessibility, blockchain introduces strong cryptographic foundations that underpin voter authentication and ballot secrecy. Public-key infrastructure enables secure identification of voters without exposing their personal information. Digital signatures authenticate ballots, while cryptographic hash functions ensure that no vote can be altered once recorded. When combined with consensus algorithms such as Proof of Stake or Practical Byzantine Fault Tolerance, blockchain creates a transparent yet confidential voting environment where every action is verified and permanently recorded.

Abuidris et al. [1] further emphasized that decentralization mitigates the risks associated with centralized authorities or intermediaries who might otherwise manipulate election data. By transferring trust from a central administrator to a network-wide protocol, blockchain transforms how legitimacy is established in digital elections. The result is a self-verifying system that minimizes human intervention and reduces the potential for internal corruption or bias. Transparency is achieved not by trusting a single authority, but by allowing every participant to observe and validate the process collectively.

Nevertheless, blockchain-based electronic voting remains a developing field with numerous technical and operational challenges. Scalability is a major concern, as national elections may involve millions of transactions occurring within a limited time window. The throughput of existing blockchain systems is often insufficient for such large-scale operations. In addition, energy-intensive consensus protocols such as Proof of Work are not suitable for electoral contexts due to high computational costs and environmental concerns. Alternative models like Proof of Stake and Delegated Proof of Stake offer better efficiency but may introduce governance complexities and partial centralization. Weinstein [3] noted that any electronic voting framework must achieve a delicate balance between efficiency, accessibility, and adherence to democratic principles, and this remains an unresolved challenge for blockchain implementations.

Another critical issue is the preservation of voter privacy and resistance to coercion. While blockchain ensures data integrity, it does not automatically protect voters from social or political pressure. Ensuring that votes cannot be linked to individuals while still maintaining verifiability requires sophisticated cryptographic techniques such as homomorphic encryption, zero-knowledge proofs, and blind signatures. These methods enable accurate vote tallying without disclosing individual choices. However, their integration into large-scale systems increases complexity and computational demand, calling for further research and optimization.

User experience and accessibility are also vital for public trust and adoption. A system that is technically secure but difficult for voters to use could inadvertently lead to disenfranchisement. Therefore, blockchain-based voting must not only achieve cryptographic robustness but also ensure usability, reliability, and clear communication of procedures to the electorate. Simple interfaces, effective voter education, and accessible verification mechanisms are necessary to ensure that the system serves all citizens equally. Moreover, compliance with national data protection laws and international electoral standards remains crucial for real-world implementation.

Globally, several pilot projects have demonstrated both the promise and limitations of blockchain in electoral contexts. Estonia has integrated distributed ledger elements into its national identity management and online voting systems. Switzerland's Zug region conducted local referendums using a blockchain platform to authenticate voters and verify ballots. Private companies such as Voatz and Polys have implemented mobile applications that leverage blockchain to enable secure remote voting. Although these initiatives showcase feasibility, independent assessments reveal limitations in scalability, transparency of the verification process, and dependency on trusted hardware environments. These mixed results underscore the necessity of continuous research, public audits, and transparent evaluation methodologies to ensure accountability and reliability.

The rapid rise of blockchain research in electronic voting has led to an extensive body of work exploring diverse architectures, consensus mechanisms, and cryptographic strategies. Scholars and practitioners are now focusing on how

to reconcile privacy with transparency, decentralization with scalability, and security with usability. As Abuidris et al. [1] and Akritidis et al. [2] observed, evaluating blockchain-driven e-voting from multiple perspectives technical, legal, and social is essential to determine its real-world feasibility and sustainability. By systematically studying existing frameworks and identifying gaps, researchers can guide the next generation of voting systems toward greater inclusiveness, accountability, and resilience.

In conclusion, blockchain technology presents a transformative approach to modernizing electoral systems through its decentralized structure, immutable ledger, and cryptographic trust model. While the technology holds immense potential to enhance transparency and security, practical implementation must address unresolved challenges related to scalability, privacy, governance, and usability. The growing body of research reflects a global commitment to reimagining democratic participation in the digital age. By consolidating technological innovation with ethical and institutional safeguards, blockchain-based e-voting can contribute to a future where elections are not only efficient but also universally trusted and verifiable.

2. Methodology

This study adopts a systematic review methodology based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol to ensure a rigorous, transparent, and reproducible assessment of existing blockchain-based electronic voting systems. The PRISMA framework provides a structured process for identifying, screening, evaluating, and synthesizing relevant literature, thereby minimizing potential biases and ensuring comprehensive coverage of the research landscape. Through this approach, the study aims to consolidate current knowledge, evaluate diverse implementation frameworks, and identify technological and operational limitations that influence the development of decentralized voting systems.

The systematic review process began with the formulation of the research objective, which focuses on understanding how blockchain technology enhances the transparency, security, and integrity of electronic voting systems. A comprehensive literature search was then performed across multiple academic databases, including IEEE Xplore, ACM Digital Library, SpringerLink, ScienceDirect, and Scopus. Additional references were obtained from Google Scholar and relevant conference proceedings to ensure inclusion of the most recent and significant contributions. The search strategy employed Boolean combinations of keywords such as *"blockchain," "electronic voting," "e-voting systems," "decentralization," "cryptography,"* and *"consensus mechanisms."* These search terms were refined iteratively to maximize retrieval of relevant publications.

All retrieved records were subjected to a multi-stage screening process. The initial screening involved the removal of duplicates and clearly irrelevant papers. In the second stage, the titles and abstracts were reviewed to determine the scope, relevance, and alignment with the research objectives. The final inclusion stage involved a full-text review to verify methodological quality, conceptual depth, and empirical rigor. Only peer-reviewed articles, book chapters, and conference papers that explicitly addressed blockchain-based electronic voting or its related cryptographic and consensus mechanisms were included. Opinion articles, incomplete technical notes, and non-English publications were excluded to maintain consistency and reliability.

For each selected study, data were systematically extracted based on several key parameters. These included publication year, blockchain framework used, consensus algorithm, type of voting model (public, private, or hybrid), cryptographic techniques employed, voter authentication mechanisms, verifiability methods, scalability considerations, and overall system performance. This structured extraction enabled cross-comparison among frameworks and identification of emerging trends, challenges, and technological advancements. The extracted data were then categorized into thematic groups, such as architectural design, security and privacy, scalability and efficiency, and real-world implementation. Thematic synthesis facilitated the identification of recurring research patterns and the formulation of a conceptual taxonomy of blockchain-driven voting systems.

To ensure methodological rigor, quality assessment criteria were applied to each study. These criteria evaluated the clarity of research objectives, the appropriateness of methodology, the depth of cryptographic analysis, and the extent of experimental validation or simulation. Studies were also assessed for transparency in reporting results and acknowledgment of limitations. Where possible, data triangulation was employed by comparing findings from multiple independent sources to strengthen reliability and minimize reviewer bias. The entire review process adhered to PRISMA's four principal stages: identification, screening, eligibility, and inclusion. This structured workflow ensured that only relevant and high-quality research was incorporated into the final synthesis.

The analysis phase involved both qualitative and quantitative examination of the collected data. Qualitative analysis focused on conceptual interpretations, system design features, and theoretical frameworks, while quantitative evaluation

summarized metrics such as latency, throughput, and computational cost as reported in the literature. Comparative tables and visual representations were prepared to highlight the diversity of blockchain implementations, consensus mechanisms, and cryptographic methods across studies. This combination of analytical perspectives provided a holistic view of how blockchain technology is being utilized to enhance e-voting systems and what limitations persist across various implementations.

Furthermore, the review process maintained objectivity through independent evaluation and verification of findings. Each paper was analyzed using a standardized evaluation matrix to ensure consistency in judgment. The adherence to PRISMA standards ensured that the review remained transparent, unbiased, and replicable. By systematically organizing the existing body of knowledge, the study not only consolidates previous findings but also establishes a foundation for future empirical investigations into blockchain-enabled e-voting technologies.

In conclusion, the methodology employed in this study guarantees a balanced and comprehensive exploration of blockchain-based electronic voting systems. The use of the PRISMA protocol provides a structured and unbiased framework for analyzing a wide range of literature sources. This methodological approach supports evidence-based interpretation and enables the identification of gaps that need to be addressed to achieve secure, scalable, and transparent e-voting infrastructures suitable for modern democratic processes.

3. Blockchain - A Potential Solution For E-Voting System

Blockchain technology has emerged as one of the most promising frameworks for addressing the persistent challenges of transparency, security, and trust in electronic voting systems. The limitations of conventional e-voting platforms such as single-point failures, tampering risks, and lack of verifiability - can be effectively mitigated by blockchain's distributed ledger structure, cryptographic integrity, and consensus mechanisms. Recent systematic reviews and surveys [4] – [12] have consistently highlighted blockchain's ability to enhance accountability and voter confidence by decentralizing control and ensuring immutability of electoral data.

Blockchain embodies several fundamental characteristics: decentralization, immutability, transparency, and cryptographic assurance. In a blockchain-based e-voting system, every vote is recorded as a transaction in a distributed ledger maintained by multiple nodes across a network. Each transaction is verified through a consensus algorithm before being permanently appended to the chain, making subsequent alteration virtually impossible. Taş and Tanrıöver [4] and Jafar et al. [5] emphasized that this decentralization eliminates dependence on a central authority and reduces the risk of insider manipulation, thereby strengthening electoral integrity. Pawlak and Poniszewska-Maranda [6] further explained that the transparency inherent in a distributed ledger enables verifiable public audits, ensuring that each recorded vote can be traced and validated without compromising voter anonymity.

Blockchain's immutability ensures that once data is recorded and confirmed by the network, it cannot be modified or deleted. This property directly addresses the integrity issues prevalent in centralized e-voting architectures. Huang et al. [7] described immutability as a technological guarantee of authenticity, where each new block cryptographically references its predecessor, forming an unbroken sequence of verified events. This immutable chain acts as a tamper-proof archive of all votes cast, establishing trust in the legitimacy of election outcomes. Moreover, Jafar et al. [8] proposed that the integration of blockchain with artificial intelligence and IoT technologies the "Secure-Tech Triad" could further enhance transparency through automated verification and anomaly detection mechanisms.

From a security perspective, blockchain ensures confidentiality, authentication, and non-repudiation by employing robust cryptographic techniques. Each voter is assigned a unique cryptographic key pair: a private key for casting the vote and a public key for verification. As Devi and Bansal [9] highlighted, this asymmetric cryptography protects voter identities while maintaining verifiable proof of participation. Benabdallah et al. [10] observed that such encryption mechanisms, coupled with digital signatures, not only validate ballot authenticity but also prevent duplication or modification during transmission. These features collectively establish a secure voting environment that minimizes the risks of fraud, double voting, and external interference.

Scalability and system performance remain crucial considerations for large-scale elections. Jafar et al. [11] addressed the need for scalable blockchain frameworks capable of processing millions of voting transactions within limited timeframes, suggesting hybrid architectures that combine private and public ledgers to balance throughput with decentralization. Vladucu et al. [12] extended this analysis, underscoring the importance of selecting consensus algorithms such as Proof of Stake (PoS), Practical Byzantine Fault Tolerance (PBFT), or hybrid models that optimize speed without compromising security.

In practice, the application of blockchain to voting is not merely theoretical. Several nations and organizations have

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piloted blockchain-enabled elections, providing valuable insights into both the opportunities and challenges associated with implementation. Estonia's pioneering I-voting system, launched in 2005, utilizes a national digital identity infrastructure to facilitate secure online voting. Although not fully blockchain-based, it laid the groundwork for future distributed verification models [13]. Similarly, Barnes et al. [14] proposed an improved architecture leveraging blockchain to enhance Estonia's framework, focusing on integrity verification and real-time auditing. Hardwick et al. [15] later introduced a dual-blockchain model one for voter identities and another for ballots ensuring both privacy and traceability. Their work demonstrated that blockchain could enforce the "one-person-one-vote" principle while maintaining voter anonymity through selective disclosure mechanisms.

Further studies have expanded upon these foundational implementations. Dreier and Sánchez [16] performed an analytical assessment of blockchain-based voting systems, emphasizing the balance between openness and confidentiality. Chen et al. [17] advanced this concept by introducing a secure digital voting system that leverages blockchain for end-to-end verification, providing resistance against double voting and post-election disputes. Ahluwalia and Kumar [18] reviewed emerging challenges in deploying blockchain for national elections, pointing out that while transparency and auditability are enhanced, issues of scalability, regulatory compliance, and voter education remain open research domains. Pereira et al. [19] contributed to this discourse by highlighting that the blockchain ledger serves as a durable and auditable record, capable of resolving disputes and ensuring accountability. Finally, Dreier and Sánchez [20] reiterated that the immutable ledger functions as a permanent archive of democratic participation, providing future generations with an incorruptible record of electoral history.

The decentralized and immutable nature of blockchain therefore directly fulfills several non-functional requirements of an ideal e-voting system namely reliability, scalability, and transparency. Decentralization distributes trust across the network, eliminating single points of failure; immutability guarantees that recorded votes remain unaltered; and transparency enables public verification while cryptographic safeguards ensure confidentiality. Together, these attributes establish a system that is not only tamper-resistant but also auditable and publicly accountable. As a result, blockchain aligns with the broader goal of reinforcing democratic trust through verifiable digital governance.

Figure 1 illustrates the essential requirements for a secure and inclusive e-voting system, integrating both functional and non-functional parameters such as universality, accessibility, reliability, and transparency. Figure 2 delineates the key properties of blockchain immutability, decentralization, distribution, and security that collectively fulfill these parameters. The synergy between the requirements of digital democracy and the inherent features of blockchain underscores its transformative potential for future electoral processes. By enabling a decentralized, transparent, and tamper-proof framework, blockchain technology lays the groundwork for trustworthy, efficient, and globally verifiable voting systems that can strengthen public confidence in modern democratic institutions.

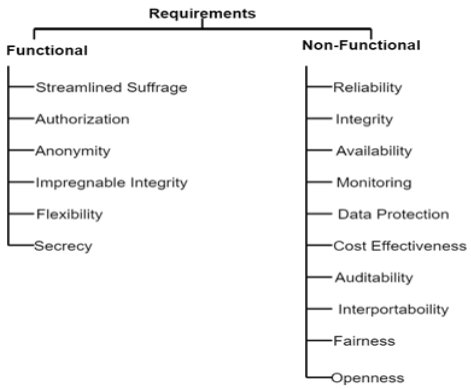

Figure 1: The requirements for a stained e-voting system

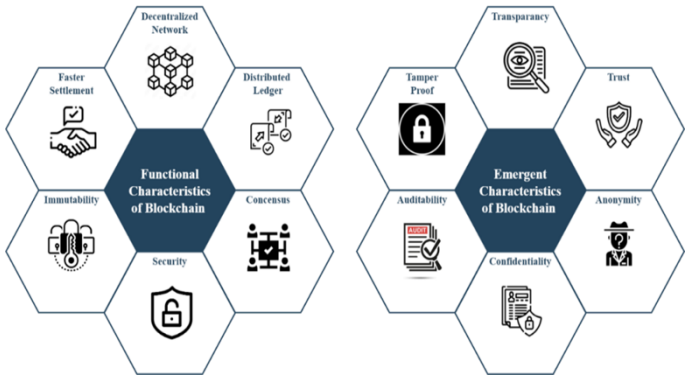

Figure 2: Blockchain Characteristics

4. Literature Review

Ahubele et al. [29] present a systematically designed system featuring a dual-layer encryption approach utilizing both Symmetric and Public key encryption, employing RSA-KEM for enhanced security.Figure 3 shows the relative analysis of the mentioned architecture based on four parameters. Electronic voting (e-voting) has evolved as a digital alternative to traditional paper-based elections, aiming to improve accessibility, transparency, and efficiency in the democratic process. By digitalizing voting procedures, e-voting systems reduce human error, accelerate vote counting, and expand participation among citizens who might otherwise face logistical barriers. However, centralized infrastructures often introduce significant vulnerabilities, including data manipulation, denial-of-service attacks, and unauthorized administrative access. The integration of blockchain technology has been increasingly recognized as a solution to these

shortcomings, as it provides decentralization, immutability, and verifiability properties essential for fostering trust and integrity in electoral systems [13]–[15].

Estonia's I-voting system, introduced in 2005, marked one of the earliest national-level implementations of remote electronic voting. It utilized national ID cards for secure voter authentication but faced challenges related to transparency, susceptibility to insider threats, and the lack of a distributed audit trail [14]. To address these vulnerabilities, Hardwick et al. [15] proposed a blockchain-based model featuring two distinct ledgers: one for voter registration and another for ballot management. In their system, voter registration is executed as a blockchain transaction verified by cryptographic credentials, while vote casting and tallying occur on a separate ledger to ensure anonymity and integrity. This dual-chain architecture enhances both privacy and verifiability but introduces scalability and latency constraints when deployed on a large scale. Consequently, later research has emphasized the importance of balancing strong security measures with operational feasibility to ensure practical applicability in national elections.

Building upon these foundational systems, numerous scholars have proposed enhanced frameworks integrating cryptographic mechanisms, distributed consensus, and smart contract functionality. Anita et al. [21] designed an improved blockchain e-voting framework incorporating a two-step voter authentication process, ensuring robust identity verification and prevention of double voting. Friðrik et al. [22] introduced a smart contract–based algorithm that automates vote validation and counting without requiring centralized supervision, thereby improving both transparency and efficiency. Similarly, Aruna et al. [23] presented an architecture combining blockchain with Merkle Root and SHA3 hashing techniques to guarantee data integrity across all stages of the voting process. Mehboob et al. [24] proposed a layered blockchain architecture that diverges from linear transactional designs, enabling modular control over different components such as voter authentication, ballot submission, and result verification. Ahmed et al. [25] introduced a blockchain-driven e-voting system emphasizing decentralized validation and dynamic verifiability, while Yi [26] explored a peer-to-peer (P2P) blockchain environment designed to remove intermediary dependence, enhancing both transparency and user trust.

Benny et al. [27] developed a blockchain-enabled platform using smart contracts to oversee the entire voting lifecycle, including ballot issuance, validation, and result compilation. Berenjestanaki et al. [28] proposed a complex privacy-preserving e-voting protocol that ensures voter anonymity and democratic integrity through the combination of multi-signature schemes and zero-knowledge proofs. Ahubele et al. [29] advanced this line of work by designing a system that employs a dual-layer encryption approach integrating both symmetric and public-key cryptography through the RSA Key Encapsulation Mechanism (RSA-KEM), thereby enhancing the confidentiality and resilience of transmitted data. This approach not only ensures privacy and verifiability but also addresses performance concerns by reducing computational load through efficient key management.

Comparative analyses of these models have identified four critical performance parameters that define the efficiency of blockchain-based e-voting systems: security, settlement time, scalability, and transparency. Figure 3 presents a relative assessment of selected architectures along these dimensions. Systems emphasizing layered encryption and distributed consensus tend to offer superior transparency and auditability, although they often exhibit slower settlement times due to computational overhead and network synchronization requirements. Nevertheless, this trade-off is generally considered acceptable in exchange for greater electoral trust and tamper resistance.

Further research has explored region-specific and domain-specific blockchain voting implementations that build upon earlier models. Roh and Lee [41] engineered an electronic voting system employing the Practical Byzantine Fault Tolerance (PBFT) consensus mechanism within a private permissioned blockchain, offering improved fault tolerance and rapid validation. Sadia et al. [40] introduced a hybrid model integrating biometrics and blockchain to authenticate voters and ensure non-repudiation. Waheed et al. [43] developed an anonymous signcryption protocol based on Elliptic Curve Cryptography (ECC), which combines confidentiality and authentication into a unified operation for optimized performance. Jumaa and Shakir [44] presented an Iraqi e-voting system utilizing lightweight ECC on a public blockchain, prioritizing security with minimal computational complexity. Jain et al. [42] proposed *MATDAAN*, an Ethereum-based e-voting scheme designed to maintain voter anonymity through the use of smart contracts, while Niloy et al. [45] introduced a United States–based blockchain voting model emphasizing transparency and integration with artificial intelligence for enhanced voter validation. Suman et al. [46] contributed the *ECC-EXONUM-eVOTING* framework, combining elliptic curve cryptography with a hybrid consensus mechanism (PBFT and RAFT) to achieve both scalability and robustness in distributed electoral environments.

Collectively, these advancements demonstrate the progression of blockchain-based voting systems from conceptual designs to practical prototypes capable of addressing multiple dimensions of electoral security and efficiency. The evolution of cryptographic methods from RSA-KEM and SHA3 to modern ECC-based schemes and zero-knowledge proofs illustrates a clear trend toward systems that reconcile voter anonymity with end-to-end verifiability. These innovations reveal that blockchain technology not only enhances transparency and security but also establishes a digital

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trust infrastructure aligned with the core principles of democratic participation.

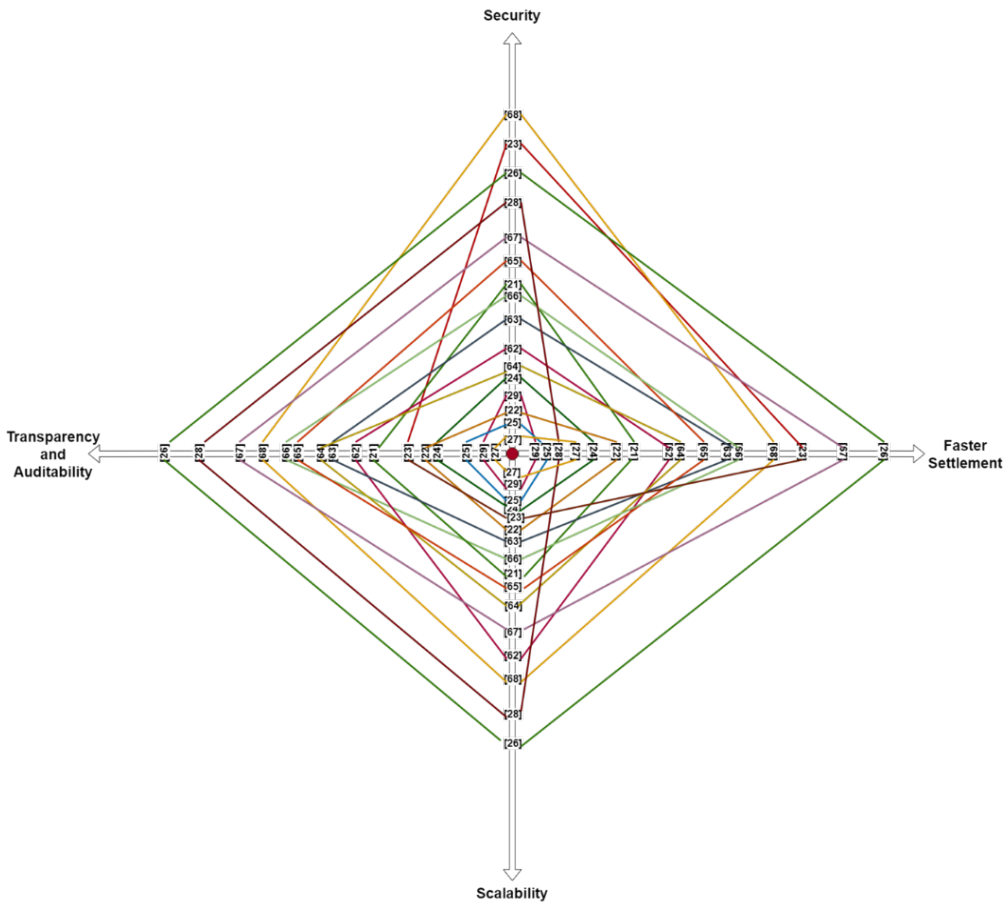

Figure 3: Relative analysis of the discussed architectures based on security, settlement time, scalability, and transparency.

The design of voter authorization mechanisms is central to ensuring both security and usability in blockchain e-voting. Eligibility verification guarantees that only registered and authenticated voters can participate, maintaining electoral legitimacy [16]. The prevention of double voting upholds the integrity of the one-person-one-vote rule [19], while the integration of anonymity-preserving access control mechanisms ensures that voter privacy remains intact throughout the election process [18]. These mechanisms form the foundation upon which the reliability and fairness of blockchain e-voting depend.

In addition to authorization mechanisms, encryption and hashing algorithms play a pivotal role in guaranteeing data integrity, confidentiality, and verifiability. Table 1 summarizes various voter authentication methods employed by researchers, while Table 2 provides an overview of encryption and hashing algorithms adopted across different blockchain e-voting architectures. The predominance of techniques such as SHA-256, SHA3, RSA, and ECC underscores their importance in enabling secure transaction validation and auditability.

Table 1: Different authentication methods used by authors

Authentication Method	Reference
ID Proof Document	[30],[31]
PH Number	[32]
Voter's Credentials	[31],[33]
Biometric verification	[15],[22],[26],[34]

Table 2: Different Hash or Encryption Algorithms used by authors

Hash/Encryption Algorithm	Reference
Homomorphic	[30],[31],[33],[35],[36],[37],[38],[39], [40],[41],[42],[
Zero Knowledge Proof	[30],[36],[37],[38],[42],[46],[47], [48],[49],[50],[51],[52],
Blind and Ring signature	[15],[28],[35],[37],[39],[45],[48], [53],[54],[55],[56],[57]
SHA 256	[26],[58],[31],[34],[35]
Keccac 256	[15],[59],[30],[35],[46],
ECC	[43],[44],[45]

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Table 3: Advantages and limitations of different proposed approaches concerning underlying strategy

Ref.	Strategy	Upsides	Downsides
[15]	Blockchain as an open and visible voting system	Adheres to the fundamental e-voting principles and incorporates decentralization; allows voters to revise ballots within the permitted timeframe	Does not provide auditability
[17]	Prêt-à-Voter e-voting method	Ensures secure end-to-end verification	Vulnerable to double voting
[26]	Separate model for synchronization, credentials, and withdrawal	Provides voter privacy	Unable to resist countermeasure attacks
[28]	Elimination of dependable intermediary	Improved security	Limited robustness
[33]	Combination of secret sharing and homomorphic encryption	Preserves anonymity	Fairness is compromised
[35]	End-to-end privacy protocol	Provides detectability	Prone to inconsistent system states
[45]	Homomorphic ElGamal encryption	Maintains anonymity	Not robust
[53]	Quantum-secured distributed ledger	Ensures concealed identity, irrevocable commitment, one-time use, confirmability, impartiality, and auto-calculation	

5. Comparative Analysis of Blockchain Frameworks for E-Voting

The choice of blockchain framework significantly influences the design, scalability, and security of electronic voting systems. Different frameworks exhibit unique strengths and limitations depending on their architectural orientation, consensus mechanism, and community support. Ethereum remains one of the most widely adopted public blockchain platforms, offering strong decentralization and smart contract capabilities. However, it suffers from scalability issues and slower transaction throughput during high network congestion. Hyperledger Fabric, in contrast, provides a permissioned blockchain architecture emphasizing privacy, modularity, and configurability, though it often relies on semi-centralized control through certificate authorities. Exonum, a lesser-known yet efficient framework, delivers high privacy and scalability through its hybrid cryptographic structure but suffers from a relatively small developer ecosystem. Quorum, an enterprise-grade Ethereum variant, offers privacy-focused smart contracts and high performance but lacks widespread adoption and suffers from anonymity challenges. Bitcoin-based frameworks emphasize decentralization and transparency but remain constrained by low transaction speed and high volatility. Custom blockchain architectures, while tailored to specific election requirements, demand substantial development resources, making standardization and interoperability difficult [11], [22], [54]–[56].

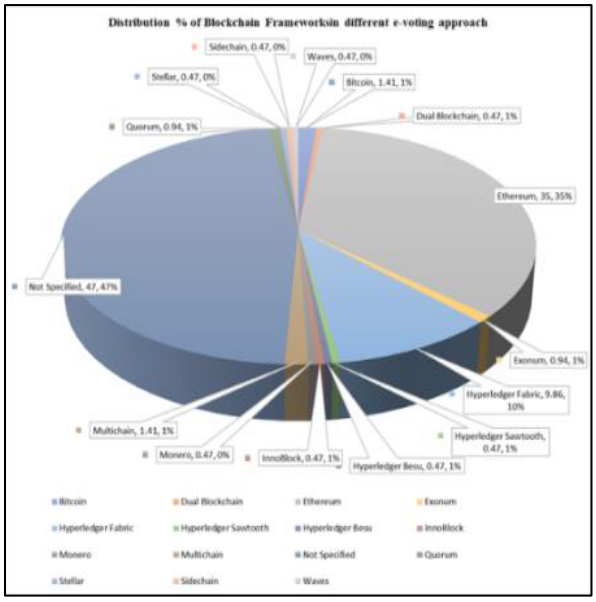

Figure 4: Different blockchain frameworks and their percentage of distribution in e-voting

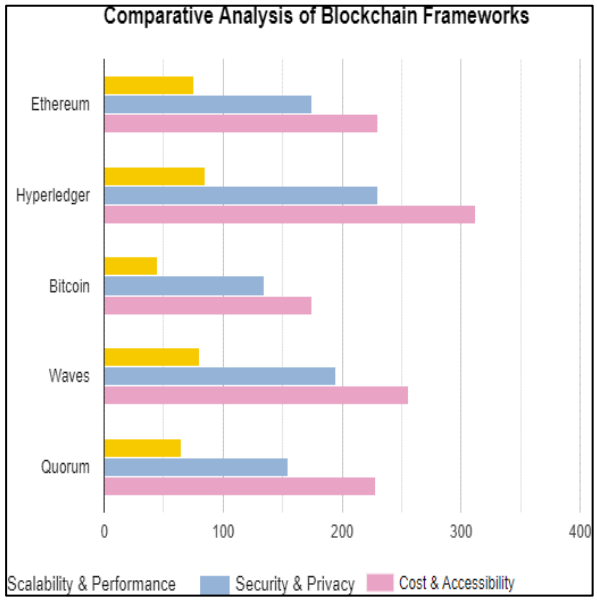

Figure 5: Relative analysis of different blockchain frameworks based on selected parameters

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Table 4: Comparative study of different blockchain-associated e-voting architectures

Ref.	Year	Framework	Language	Cryptographic Protocol	Consensus Algorithm	Critical Design Decision / Emphasis	Enhancement Opportunities / Constraints
[68]	2023	AVISPA simulator	Python 2.7.18	ECC	PBFT and RAFT	Electronic voting on Exonum private blockchain; robust elliptic curve cryptography; hybrid consensus (PBFT and RAFT); ECDH-generated keys; validator-managed key entities; lightweight DApp for space-limited devices	Secure short-duration transactions; ECC-EXONUM-eVOTING offers efficient, secure e-voting
[67]	2023	Ganache	Solidity, Rust, JavaScript, Vyper	Not mentioned	Proof of Stake	U.S. e-voting system emphasizing security and transparency; reusable smart contracts; integration with Web 3.0; AI-based authentication; SQL database for voter data	Secure but with higher storage costs and operational overheads
[13]	2022	Not specified	Not specified	RSA-KEM	Not specified	Utilizes symmetric and public-key encryption; private key wrapping scheme; key generation and management techniques	RSA-KEM efficient but limited against quantum attacks
[21]	2020	.NET with MS SQL Server 2008	C#	SHA-256	Not specified	User access via OTP; vote casting and result calculation	Unable to prevent double voting; weak security
[23]	2020	Django	Python	SHA3 and Merkle Root	Not specified	Biometric voter verification; post-election vote tracking; Merkle tree for data integrity	Needs parallelized transactions for scalability
[25]	2020	Django 2.1.2 (PyCharm IDE)	Python 3.7	SHA-256	Not specified	Private key ballot signing; ballot sealing via mining	Public chains vulnerable to high turnout; privacy-audit trade-off
[26]	2019	Not specified	Not specified	SHA-256	Not specified	ECC for unique voter signatures; comprehensive block verification	Potential PKI vulnerabilities may compromise integrity
[47]	2018	Hyperledger Fabric	Not specified	Proof-of-Knowledge	Byzantine Fault Tolerance	Compact linkable ring signatures within Hyperledger Fabric	Suggests Hyperledger Sawtooth for parallel execution
[37]	2018	Not specified	Not specified	SHA-1	Not specified	Universal e-voting secured with SHA; vote blocks linked to candidate chains	Complex multi-chain structure increases overhead
[48]	2018	Exonium, Quorum, Geth	Not specified	Not specified	Not specified	District nodes manage boot node smart contracts	Exonium costly for scale; Quorum and Geth lack parallel support
[49]	2018	Not specified	Not specified	Not specified	Not specified	Multi-chain architecture; TTP authenticates voters	Delay in secret message verification
[50]	2017	Not specified	Not specified	SHA-1	Not specified	Hierarchical voting architecture using dual blockchains	Suggests Hyperledger Sawtooth for scalability
[51]	2017	Not specified	Not specified	SHA-256	Longest Chain Rule	Blockchain-based model for Estonia's I-Voting	High storage and processing demands
[52]	2017	Not specified	Not specified	SHA-256	PoW and PoS	Blind signatures enable secure vote validation	High latency in large-scale voting

https://doi.org/10.5281/zenodo.17565011

Figure 4 illustrates the percentage distribution of various blockchain frameworks utilized in e-voting systems, reflecting Ethereum's dominance in academic and prototype implementations. Hyperledger and Exonum demonstrate growing adoption in institutional and governmental projects, while emerging frameworks like Quorum and Waves show potential in permissioned or low-cost voting environments. Figure 5 presents a relative comparative analysis of blockchain frameworks based on selected technical parameters such as performance, scalability, transparency, and privacy. Table 5 provides a detailed summary of the major frameworks, highlighting their respective advantages and limitations.

Table 5. Comparative Study of Different Blockchain Frameworks

Framework	Upsides	Downsides
Ethereum	Reliability; high market capitalization; extensive developer community	Slow transaction speed; scalability limitations; partial centralization tendencies
Hyperledger Fabric	Modular architecture; scalability; hybrid deployment models with fine-grained permissions	Complex architecture; limited fault tolerance; reliance on centralized certificate authority
Bitcoin	High transparency; capitalized ecosystem; low transaction cost	Volatility; low transaction throughput; unsuitable for real-time elections
Waves	High accessibility; low operational cost; scalable lightweight architecture	Limited community support; performance inconsistency under heavy load
Quorum	Enhanced privacy; high throughput; voting-based consensus	Limited anonymity; transaction verification by selected nodes; restricted block size; messaging overhead

6. Blockchain-Based Projects Implemented or Under Development

These projects collectively underscore blockchain's potential to redefine electoral trust through decentralization, transparency, and cryptographic assurance. However, they also reveal persistent challenges in scalability, usability, and compliance, indicating the need for standardization and continued empirical validation across diverse geopolitical contexts. A number of blockchain-based electronic voting initiatives have emerged globally, reflecting growing confidence in distributed ledger technology as a foundation for secure and transparent electoral systems. These projects vary in their technical design, cryptographic structure, and deployment scale, but they share a common vision of establishing tamper-resistant and trust-enhancing voting mechanisms.

Oluwafemi et al. [56] proposed a blockchain-oriented framework for e-voting that emphasizes decentralization, data integrity, and verifiability. Their study introduced cryptographic validation and consensus-based recordkeeping to eliminate single points of failure and prevent unauthorized modifications. Adhikari et al. [57] presented a insights into how generative and analytical frameworks can assist in evaluating emerging digital systems. Although focused on AI-driven content generation, their methodological approach to comparing models and interpretability metrics provides a valuable analytical basis for assessing blockchain-based systems.

Banerjee and Saha [58] investigated blockchain signatures as a mechanism to ensure information integrity and non-repudiation, illustrating how cryptographic proofs can safeguard voting records against tampering. Their study demonstrated that blockchain-based authentication significantly enhances the traceability and security of digital transactions an essential requirement for online voting environments. In a subsequent study, Banerjee and Saha [59] analyzed smart contract–based blockchain applications, highlighting their relevance to e-voting by enabling decentralized logic, automated tallying, and transparent audit trails. Das and Bora [60] focused on IoT data clustering frameworks, emphasizing the role of distributed analytics and consensus-driven validation. Their findings extend naturally to blockchain ecosystems, where scalable, node-based coordination is critical for handling large volumes of voter and transactional data.

Together, these research initiatives reflect the interdisciplinary evolution of blockchain-based voting bridging cryptography, distributed systems, and intelligent data analysis. Each contribution demonstrates a distinct perspective: from structural decentralization [56], to computational model comparison [57], to secure transaction integrity [58], to the automation of consensus protocols [59], and finally to scalable data validation [60] – [62]. Collectively, they underscore blockchain's transformative role in ensuring transparency, privacy, and accountability in electronic voting systems.

However, despite these advancements, challenges persist. The practical deployment of blockchain voting platforms remains limited by scalability constraints, heterogeneous infrastructure requirements, and the need for regulatory harmonization across jurisdictions. Future research must therefore focus on empirical validation, user-centered design, and cross-platform interoperability to transform these academic frameworks into viable real-world systems capable of supporting large-scale democratic participation.

7. Future Enhancements and Research Directions

The present review provides a comprehensive understanding of the evolution, advantages, and limitations of blockchain-based e-voting systems. Despite substantial theoretical progress, there remains a noticeable lack of large-scale empirical evaluations assessing performance, reliability, and user acceptance. Future research must address these gaps through multidimensional approaches that combine technical, social, and regulatory analyses.

Empirical Validation of Proposed Solutions: Although numerous blockchain voting frameworks have been proposed, few have undergone systematic real-world testing. Implementing these models in controlled pilot elections will provide quantitative insights into scalability, latency, and user behavior.

Comprehensive Examination of Existing Projects: While several blockchain-based projects are discussed in this study, future research should focus on detailed comparative assessments of their outcomes, challenges, and long-term sustainability to guide future implementations.

User Experience and Adoption: Beyond technical feasibility, user acceptance remains crucial for widespread adoption. Future work should explore user perception, ease of use, and trust in blockchain voting systems to ensure inclusivity across demographics.

Security and Privacy in Practical Scenarios: Although cryptographic privacy mechanisms have been established theoretically, practical implementations must evolve to withstand emerging threats, particularly in public and hybrid blockchain networks.

Evolving Blockchain Ecosystem: With the rapid advancement of consensus algorithms and interoperability frameworks, continuous evaluation of new blockchain technologies is essential to maintain the integrity and adaptability of e-voting solutions.

8. Conclusion

This comprehensive survey of blockchain-integrated electronic voting systems highlights the transformative potential of distributed ledger technology in modern electoral processes. By addressing the inherent limitations of traditional and centralized voting architectures, blockchain introduces verifiable transparency, data integrity, and citizen trust. The comparative analysis of blockchain frameworks demonstrates that while each platform offers distinct advantages, no single architecture fully satisfies all functional and non-functional requirements. Real-world implementations from Luxoft's Hyperledger-based system in Switzerland to Ethereum-powered solutions such as Polys and DecentraVote illustrate the growing maturity of blockchain in civic applications.

In conclusion, blockchain technology represents a paradigm shift in digital democracy, offering a foundation for transparent, tamper-resistant, and inclusive voting systems. As future research expands empirical testing and usability assessment, the convergence of blockchain with privacy-preserving cryptography, artificial

intelligence, and secure IoT integration promises to revolutionize how democratic participation is conducted in the digital era.

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